ONTAP clusters

NetApp Keystone

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ONTAP clusters

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`GET`	`/v2.1/ontapclusters/{id}`	Retrieve an ONTAP cluster by ID.
`POST`	`/v2.1/ontapclusters`	Create a new ONTAP cluster.
`PUT`	`/v2.1/ontapclusters/{id}`	Update ONTAP cluster inventory by ID.
`DELETE`	`/v2.1/ontapclusters/{id}`	Delete an ONTAP cluster.

ONTAP clusters attributes

The following table lists the ONTAP cluster attributes.

Attribute	Type	Description
`id`	String	The unique identifier of the ONTAP cluster.
`name`	String	The ONTAP cluster name.
`uuid`	String	The ONTAP cluster universal unique identifier (UUID).
`management_ip`	String	The ONTAP cluster management IPv4 address.
`username`	String	The ONTAP cluster name
`password`	String	The ONTAP cluster password
`provisioning_state`	String	Identifies whether a cluster is available for provisioning operations. The options include: • Open • Closed
`data_network_ip_cidr`	String	The CIDR notation of a subnet.
`data_network_default_gateway`	String	The IPV4 address.

Attribute	Type	Description

`data_network_ports`	–	A list of the ONTAP cluster data network ports.
		
		``
		For example:
		[
		  {
		  "node_name": "dev-ots-per01-01",
		  "port_name": "e0c-120",
		  "parent_port": "e0c"
		  }
		]
		``

`intercluster_lifs`	–	The ONTAP cluster intercluster LIFs.
		
		``
		For example:
		[
		  {
		  "name": "peer1",
		  "node": "aff-01",
		  "port": "a0a-103",
		  "address": "10.128.113.232",
		  "netmask": "255.255.255.0"
		  },
		{
		  "name": "peer2",
		  "node": "aff-02",
		  "port": "a0a-103",
		  "address": "10.128.113.233",
		  "netmask": "255.255.255.0"
		  }
		]
		``

Attribute	Type	Description
`svm_root_service_level`	String	The ONTAP cluster storage virtual machine (SVM) root service level name. Applicable values are Standard, Extreme, or Premium. This service level is assigned by default to all SVMs created under the cluster. The cluster should have an associated aggregate for the service level mentioned.
`zone`	String	The zone name.
`Subscription_number`	String	The Zuora subscription.
`services_available`	–	List of services available and their state. For example: `{ "fcp": false, "iscsi": true, "nas": true }`
`data_fcp_ports`	–	List of nodes and ports for FCP-enabled ONTAP clusters. For example: `[ { "node_name": "aff-01", "port_name": "0g" }, { "node_name": "aff-01", "port_name": "0h" ]`
`is_mcc`	Boolean	Indicates whether the cluster is MetroCluster enabled or not. The default is False.
`mcc_partner_cluster`	String	The identifier of the partner cluster of the current cluster in a MetroCluster pair. Required if the cluster is enabled for MetroCluster.

Retrieve all ONTAP clusters

Use the method listed in the following to retrieve all ONTAP clusters or a subset of ONTAP clusters.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/ontapclusters`	Retrieve all ONTAP clusters.	`offset` and `limit` – see Common Pagination

Required request body attributes: `none`

Request body example:

```
none
```

Response body example:


```
{
  "status": {
    "user_message": "Okay. Returned 2 records.",
    "verbose_message": "",
    "code": 200
  },
  "result": {
    "returned_records": 2,
    "total_records": 5,
    "sort_by": "created",
    "order_by": "desc",
    "offset": 3,
    "limit": 2,
    "records": [
      {
        "id": "5c5bb9f16680a7002a5f7450",
        "name": "dev-ots-per01",
        "region": "au-west1",
        "zone": "au-west1-a",
        "uuid": "63053baa-ada4-11ea-b197-005056a4c0ef",
        "management_ip": "10.128.115.173",
        "username": "admin",
        "services_available": {
          "fcp": false,
          "iscsi": true,
          "nas": true
        },
        "provisioning_state": "open",
        "data_network_ports": [
```

```
          {
            "node_name": "dev-ots-per01-01",
            "port_name": "e0c-120",
            "parent_port": "e0c"
          }
        ],
        "data_network_ip_cidr": "10.96.120.0/24",
        "data_network_default_gateway": "10.96.120.1",
        "svm_root_service_level": "performance",
        "intercluster_lifs": [
          {
            "name": "dev-ots-per01-01-icl01",
            "node": "dev-ots-per01-01",
            "port": "e0b",
            "address": "10.128.115.144",
            "netmask": "255.255.255.0"
          }
        ],
        "subscription_number": "A-S00003875",
        "created": "2019-02-22T03:38:38.867Z",
        "data_fcp_ports": []
      },
      {
        "id": "5eaf5249f038943eb46b6608",
        "name": "aff",
        "region": "au-east1",
        "zone": "au-east1-b",
        "uuid": "62d649d2-07a1-11e6-9549-00a0985c0dcb",
        "management_ip": "10.128.113.69",
        "username": "admin",
        "services_available": {
          "fcp": true,
          "iscsi": true,
          "nas": true
        },
        "provisioning_state": "open",
        "data_network_ports": [
          {
            "node_name": "aff-01",
            "port_name": "a0a-2000",
            "parent_port": "a0a"
          },
          {
            "node_name": "aff-02",
            "port_name": "a0a-2000",
            "parent_port": "a0a"
```

```
          }
        ],
        "data_network_ip_cidr": "10.50.50.0/24",
        "data_network_default_gateway": "10.50.50.1",
        "svm_root_service_level": "premium",
        "intercluster_lifs": [
          {
            "name": "peer1",
            "node": "aff-01",
            "port": "a0a-103",
            "address": "10.128.113.232",
            "netmask": "255.255.255.0"
          },
          {
            "name": "peer2",
            "node": "aff-02",
            "port": "a0a-103",
            "address": "10.128.113.233",
            "netmask": "255.255.255.0"
          }
        ],
        "subscription_number": "A-S00004635",
        "created": "2019-02-22T03:38:38.867Z",
        "data_fcp_ports": [
          {
            "node_name": "aff-01",
            "port_name": "0g"
          },
          {
            "node_name": "aff-01",
            "port_name": "0h"
          },
          {
            "node_name": "aff-02",
            "port_name": "0g"
          },
          {
            "node_name": "aff-02",
            "port_name": "0h"
          }
        ],
        "is_mcc": false,
        "created": "1995-09-07T10:40:52Z"
      }
    ]
  }
```

```
}
```

Retrieve ONTAP clusters by ID

Use the method listed in the following to retrieve ONTAP clusters by ID.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/ontapclusters/{id}`	Retrieve the ONTAP clusters identified by ID.	`id (string)`: The unique identifier of ONTAP cluster.

Required request body attributes: `none`

Request body example:

```
none
```

Response body example:


```
{
  "status": {
    "user_message": "Okay. Returned 1 record.",
    "verbose_message": "",
    "code": 200
  },
  "result": {
    "returned_records": 1,
    "records": [
      {
        "id": "5c5bb9f16680a7002a5f7450",
        "name": "dev-ots-per01",
        "region": "au-west1",
        "zone": "au-west1-a",
        "uuid": "63053baa-ada4-11ea-b197-005056a4c0ef",
        "management_ip": "10.128.115.173",
        "username": "admin",
        "services_available": {
          "fcp": false,
          "iscsi": true,
          "nas": true
        },
        "provisioning_state": "open",
        "data_network_ports": [
          {
            "node_name": "dev-ots-per01-01",
```

```
            "port_name": "e0c-120",
            "parent_port": "e0c"
          }
        ],
        "data_network_ip_cidr": "10.96.120.0/24",
        "data_network_default_gateway": "10.96.120.1",
        "svm_root_service_level": "performance",
        "intercluster_lifs": [
          {
            "name": "dev-ots-per01-01-icl01",
            "node": "dev-ots-per01-01",
            "port": "e0b",
            "address": "10.128.115.144",
            "netmask": "255.255.255.0"
          }
        ],
        "subscription_number": "A-S00003875",
        "created": "2019-02-22T03:38:38.867Z",
        "data_fcp_ports": [],
        "is_mcc": false,
        "created": "1995-09-07T10:40:52Z"
      }
    ]
  }
}
```


Create ONTAP cluster

Use the API in the following table to create an ONTAP cluster.

ONTAP clusters are always created with the iSCSI service enabled. Optionally, the FCP service can be enabled if the infrastructure supports it.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/ontapclusters`	Create an ONTAP cluster.	None

Required request body attributes: `name`, `uuid`, `management_ip`, `username`, `password`, `data_network_ip_cidr`, `data_network_default_gateway`, `intercluster_lifs`, `zone`

If FCP is enabled (using the `services_available` FCP attribute), the `data_fcp_ports `are required.

If `is_mcc` is true, the `mcc_partner_cluster` is required.

Request body example:

```
{
  "name": "clustername",
  "uuid": "49b6e08e-513a-11ea-b197-005056a4c0ef",
  "management_ip": "10.128.112.165",
  "username": "admin",
  "password": "ClusterPassword",
  "provisioning_state": "open",
  "data_network_ip_cidr": "10.96.112.0/24",
  "data_network_default_gateway": "10.96.112.1",
  "data_network_ports": [
    {
      "node_name": "clustername-01",
      "port_name": "e0c-112",
      "parent_port": "e0c"
    }
  ],
  "intercluster_lifs": [
    {
      "name": "clustername-01-icl01",
      "node": "clustername-01",
      "port": "e0b",
      "address": "10.128.112.222",
      "netmask": "255.255.255.0"
    }
  ],
  "svm_root_service_level": "extreme",
  "zone": "MyZone",
  "subscription_number": "",
  "services_available": {
    "fcp": false,
    "iscsi": true,
    "nas": true
  },
  "data_fcp_ports": [
  ],
  "is_mcc": false,
}
```


Response body example:


```
{
    "status": {
        "user_message": "Okay. New resource created.",
        "verbose_message": "",
        "code": 201
```

```
    },
    "result": {
        "returned_records": 1,
        "records": [
            {
                "id": "5ef155b8f5591100010a75c5",
                "name": "clustername",
                "region": "MyRegion",
                "zone": "MyZone",
                "uuid": "49b6e08e-513a-11ea-b197-005056a4c0ef",
                "management_ip": "10.128.112.165",
                "username": "admin",
                "services_available": {
                    "fcp": false,
                    "iscsi": true,
                    "nas": true
                },
                "provisioning_state": "open",
                "data_network_ports": [
                    {
                        "node_name": "clustername-01",
                        "port_name": "e0c-112",
                        "parent_port": "e0c"
                    }
                ],
                "data_network_ip_cidr": "10.96.112.0/24",
                "data_network_default_gateway": "10.96.112.1",
                "svm_root_service_level": "extreme",
                "intercluster_lifs": [
                    {
                        "name": "clustername-01-icl01",
                        "node": "clustername-01",
                        "port": "e0b",
                        "address": "10.128.112.222",
                        "netmask": "255.255.255.0"
                    }
                ],
                "subscription_number": "",
                "created": "2020-06-23T01:07:04.563Z",
                "data_fcp_ports": [],
                "is_mcc": false,
                "mcc_partner_cluster": "5d2fb0fb4f47df00015274e3",
                "created": "1995-09-07T10:40:52Z"
            }
        ]
    }
```

```
}
```

Modify ONTAP cluster

Use the method listed in the following to modify the ONTAP cluster.

HTTP Method	Path	Description	Parameters
`PUT`	`/v2.1/ontapclusters/{id}`	Modify the details of the ONTAP cluster identified by ID.	`id (string)`: The unique identifier of ONTAP cluster.

Required request body attributes: `none`

Request body example:

```
{
  "name": "clustername",
  "uuid": "49b6e08e-513a-11ea-b197-005056a4c0ef",
  "management_ip": "10.128.112.165",
  "username": "admin",
  "password": "ClusterPassword",
  "provisioning_state": "open",
  "data_network_ip_cidr": "10.96.112.0/24",
  "data_network_default_gateway": "10.96.112.1",
  "data_network_ports": [
    {
      "node_name": "dev-ots-syd01-01",
      "port_name": "e0c-112",
      "parent_port": "e0c"
    }
  ],
  "intercluster_lifs": [
    {
      "name": "dev-ots-syd01-01-icl01",
      "node": "dev-ots-syd01-01",
      "port": "e0b",
      "address": "10.128.112.222",
      "netmask": "255.255.255.0"
    }
  ],
  "svm_root_service_level": "standard",
  "zone": "MyZone",
  "subscription_number": "",
  "services_available": {
    "fcp": false,
    "iscsi": true,
    "nas": false
  },
  "data_fcp_ports": [
  ]
}
```


Response body example:


```
{
    "status": {
        "user_message": "Okay. Accepted for processing.",
        "verbose_message": "",
        "code": 202
    },
```

```
    "result": {
        "returned_records": 1,
        "records": [
            {
                "id": "5ef155b8f5591100010a75c5",
                "name": "clustername",
                "region": "MyRegion",
                "zone": "MyZone",
                "uuid": "49b6e08e-513a-11ea-b197-005056a4c0ef",
                "management_ip": "10.128.112.165",
                "username": "admin",
                "services_available": {
                    "fcp": false,
                    "iscsi": true,
                    "nas": true
                },
                "provisioning_state": "open",
                "data_network_ports": [
                    {
                        "node_name": "dev-ots-syd01-01",
                        "port_name": "e0c-112",
                        "parent_port": "e0c"
                    }
                ],
                "data_network_ip_cidr": "10.96.112.0/24",
                "data_network_default_gateway": "10.96.112.1",
                "svm_root_service_level": "standard",
                "intercluster_lifs": [
                    {
                        "name": "dev-ots-syd01-01-icl01",
                        "node": "dev-ots-syd01-01",
                        "port": "e0b",
                        "address": "10.128.112.222",
                        "netmask": "255.255.255.0"
                    }
                ],
                "subscription_number": "",
                "created": "2020-06-23T01:07:04.563Z",
                "data_fcp_ports": [],
                "is_mcc": false,
                "mcc_partner_cluster": "5d2fb0fb4f47df00015274e3",
                "created": "1995-09-07T10:40:52Z"
            }
        ]
    }
}
```

Delete an ONTAP cluster

Use the method listed in the following table to delete an ONTAP cluster.

HTTP Method	Path	Description	Parameters
`DELETE`	`/v2.1/ontapclusters/{id}`	Delete the ONTAP cluster identified by ID.	`id (string)`: The unique identifier of the ONTAP cluster.

Required request body attributes: `none`

Request body example:

```
none
```

Response body example:

```
No content for succesful delete
```

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